

Safe System of Work

Casting Over Quarry Face

Ref. SSoW-057	Author.	Lee Hardy	Date:	03.08.2022
---------------	---------	-----------	-------	------------

Reviewed By: Name	Derick Bibby	Date of Review:	09.02.2025
-------------------	--------------	-----------------	------------

Record of Amendments			

Next Review Due Date:			09.02.2026
-----------------------	--	--	------------

Overview

The following information provides guidance on the company's Standard Operation Procedures for works involving the casting of material over a quarry face.

All employees have a responsibility for their safety and the safety of others by adhering to these codes of practice.

The Operator must always refer to the task Risk Assessment prior to starting works.

Hazards Identified

- Adverse weather
- Crushing
- Machine Rollover
- Vehicle movements
- Falls from height
- Slips & Trips

Personal Protective Equipment (PPE) Requirements



Inspection and Monitoring

Prior to the commencement of casting the working area will be inspected by a competent person and a safe method of working agreed with the operative. This will include agreed areas where casting over the face can be undertaken safely and the safe positioning of the excavator or shovel. The agreed safe casting areas should be demarcated on the ground.

The pre-work inspection will estimate the likely extent that boulders could reach on the benches below and measures will be put in place to restrict access or control the rock fall with bunding.

During casting the operator of the shovel or excavator shall constantly monitor the bench for any signs of cracking or slumping that could give rise to the failure of the working area.

Daily inspection to be undertaken to confirm stability of the blast pile, excavator platform and adequacy of stand-off restrictions.

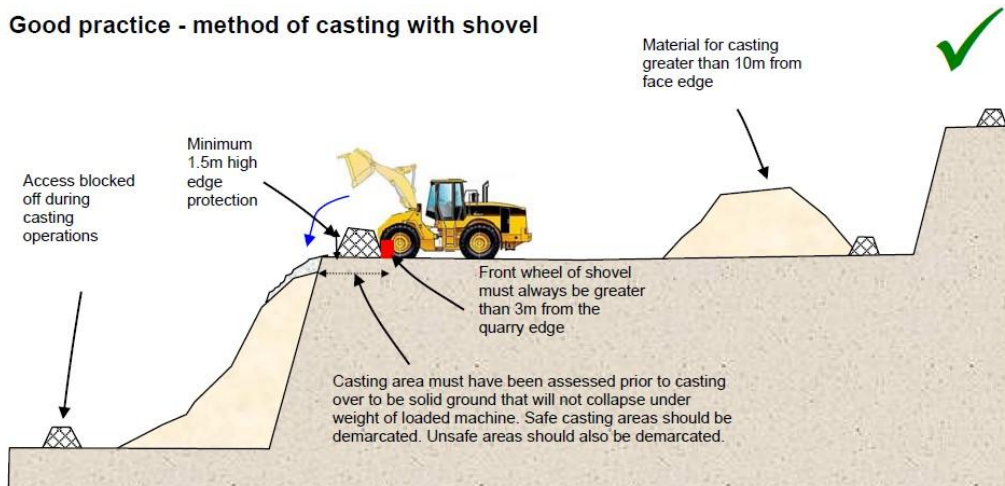
Should any signs of instability be noted, all work should be suspended, and access prohibited until further geotechnical advice is sought.

Safe Method of Operation

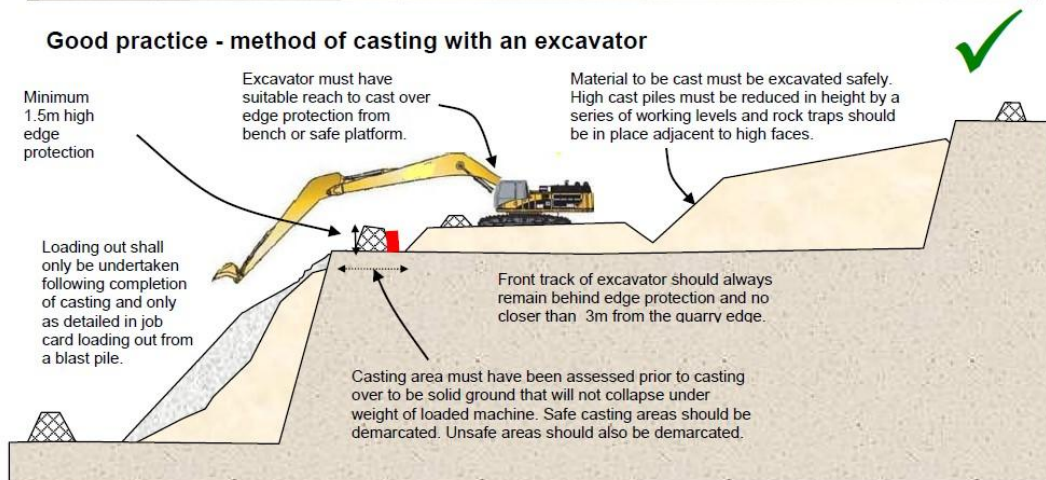
- Quarry Manager/Supervisor to brief all operatives on works to be undertaken and any permits to be signed on to.
- Communications (radios) to be tested and all operatives to confirm they are aware of call signs.
- Only a suitably sized excavator or loading shovel should be used when casting so that material can be cast over edge protection from a safe operational area. The excavator or shovel should be of sufficient weight to safely move the largest anticipated boulder. Large boulders that require pecking down should be placed to a position where safe access is possible for future pecking down.
- High cast piles must be worked in a series of working levels with progressive casting down from top to bottom. At all stages the excavator must not be at risk from rock fall from high faces above the excavator.
- The excavator must always stay on a solid bench or platform and no machine is permitted to venture on to the cast pile unless this has been assessed to be stable.
- An edge protection bund must be created at the face edge, a minimum of 3m from the face edge and 1.5m high. This will also serve as an indication of safe distance for the machine operator.
- The working area must be an exclusion zone to pedestrians and all access routes must be blocked off. If absolutely necessary to exit machine, the operator must not walk on the material to be cast or cast pile and must always maintain a safe retreat from the casting area at all times.
- During casting, the operational area below must be blocked off to prevent boulders rolling on to roadways. A bund must be created at the toe of the face, a minimum of half the face height away and 2m high. Every time the cast pile is loaded out at the base, the bund is reinstated as required.
- It is also important that any excavation of material to be cast does not undermine any access roadways.
- During casting, no working of the material cast over the face is permitted to be undertaken at the same time. Loading out can only take place when casting operations have been formally suspended.
- Care must be taken in some cast over piles as there is a greater risk of failure due to the high angled planes of weakness that can occur when casting compared to that in a blast pile. High cast piles will need to be safely cast down with an excavator prior to loading out.

- All site personnel approaching the casting working area must indicate their presence to the excavator driver and work suspended until they have left the area and it is safe to continue.
- When not in use, machines should be parked at least 10m away from above any open edge, with the cab door/opening positioned away from the quarry face.

Good practice - method of casting with shovel



Good practice - method of casting with an excavator



Bad practice - method of casting with an excavator

